LESSON - 6

MUSIC HELPS PLANTS GROW

Session - 1

22NNTG

I PRE-READING

- What are the similarities and differences between plants and humans ? Discuss with your friends and make a note of them.
- Music makes us happy. Does music have any effect on plants ? Let's read this lesson to know something about it.

II WHILE-READING

Text

- **SGP-1**
- Read paras - I to 3 silently and answer the questions that follow.

1. Many scientists believe that music soothes a plant. It also helps it grow. This is now a scientific fact.
2. Les Harsten, a sound engineer from New York, carried out some exciting experiments. Here is one of them.
3. Harsten's theory was that plants definitely react to music. In his experiment, he used two banana plants. He gave both plants the same light, heat and water. But, one of the plants 'listened' to some music for about an hour a day. This music, in fact, was a high-pitched humming sound. He found that this plant grew faster. It also grew 70 percent taller than the other plant!

- **Comprehension Questions:**

    1. The first paragraph is about a scientific fact. What is that ?
    2. What is the second paragraph about ?
    3. Who is Les Harsten ?
    4. What is his native place ?

5. What is the third paragraph about ?
6. What is Harsten's theory ?
7. What did he use for his experiment ?
8. What did he give to both the banana plants ?
9. He exposed one of the plants to music. How many hours a day ?
10. What was his finding ?
11. What happened to the plant that listened to music ?

Session - 2

- **SGP-2**
- Read silently paragraphs 4 and 5 and answer the questions that follow:

4. Lynn and Joe Rapp, the authors of a book called 'Indoor Plants', say that plants respond to all sounds, whether it is music or voices. These sound waves make vibrations which stimulate growth. They say plants have definite likes and dislikes in music- and they seem to like classical music more!

5. Harsten explained this in a very scientific way. He said the hum (or similar music) stimulated the plant's breathing cells and made them stay open for longer periods. Because of this, the plant took in more nutrients from the air than it normally would, and thus it grew faster. He also said that if the sound was played continuously, the breathing-cells would not be able to close. Therefore, the plant would grow so fast that it would kill itself.

- **Comprehension Questions:**

1. Who are the authors of - "Indoor Plants" ?
2. What do they say ?
3. What does 'this' (in the first line of para - 5) refer to ? What do you understand by 'stimulate' ?
4. What did music stimulate ? (Para - 5)
5. What happened when the breathing cells of the plant were stimulated ?
6. What happened when the cells stayed open for longer periods ?
7. What would happen if the music was played continuously ?
8. What is the fifth paragraph about ?
9. How is the finding of Lynn and Rapp different from that of Harsten ?

10. Do plants have likes and dislikes in music according to Lynn and Rapp ?
11. What type of music do plants like ?
12. Do you know any Indian scientists who have experimented on plants ? Any such a scientist of Odisha ?

Session - 3

III POST-READING

1 Visual Memory Development Technique (VMDT) :

Whole Text : Les Harsten-a sound engineer, banana plants, Indoor plants, dislike classical music, breathing cells

Part : (para- 5) scientific ways, breathing cells, more nutrients, plants would grow so fast

2 Comprehension Activities :

Answer the following questions choosing the most appropriate answer from the options.

1. 'It' in the second sentence in para-1 refers to ________________.

   A. Plant   B. music

   C. scientific fact   D. scientists

2. Les Harsten was a ______________.

   A. Civil engineer   B. sound engineer

   C. software engineer   D. electrical engineer

3. Harsten, in his experiment, gave both banana plants ________________.

   A. Light and heat   B. heat and water

   C. light and water   D. light, heat, and water

4. Which banana plant grew faster?

   A. The plant which listened to the music.

   B. The plant which got more water.

   C. The plant which got more water and light.

   D. The plant which got light, heat and water.

5. The plants can take more nutrients if their breathing cells ________________.

   A. remain closed   B. stay open   C. get water   D. get heat

Session - 4

3 Listening :

- Your teacher reads out the following statements.
  Listen to him/her and say whether they are true or false.
  (i) Lynn is a sound engineer.
  (ii) Harsten's theory says that plants definitely react to music.
  (iii) Lynn and Harsten are the writers of the book, "Indoor Plants."
  (iv) The vibration of the sound waves increases the growth of plants.
  (v) The plants don't like classical music.
  (vi) The humming sounds make the cells open for longer periods.
  (vii) The plants take in more nutrient when their cells are closed.

4 Speaking :

Chain-drill:
"Music helps the plants grow faster."
"Plants like classical music more."

Session - 5

5 Vocabulary :

Match words under 'A' with their meanings under 'B'. One has been done for you.

A	B
• soothe	• to make someone move active/alert
• hum	• very small unit of living matter
• vibration	• to cause strong feeling of eagerness/interest
• stimulate	• to make someone/something calm/quiet
• cell	• to sing a tune with one's lips closed
• excite	• continuous rapid shaking movement

6 Usage :

a. We add '-ing' to verbs and use before nouns to talk more about them; for example, 'breath+ ing=breathing' and it is used before 'cells' in the text as 'breathing cells'.

(i) Now add '–ing' to the verbs in the box-1, use them before the nouns choosing from the box-2(Refer to the text.)

Box-1	Box-2
breath	plants
grow	cells
excite	waves
hum	sounds
sooth	experiments

Example: 1. Breathing cells

2. __________ ___________

3. __________ ___________

4. __________ ___________

5. __________ ___________

(ii) Then fill in the blanks with the above suitable pairs in the sentences given below. One is done for you.

1. There are ________ ________ on plants and music.
2. The ________ ________ help plants grow fast.
3. The music stimulates the plants' breathing cells to stay open.
4. The ________ ________ activate the growth of the plants.
5. The continuous ________ ________ help plants to take more nutrients.

Session - 6

b. We combine our ideas by combining small parts together and we make a long sentence. Now complete the following sentences choosing the suitable parts from the box.

(i) Many scientists believe

____

(ii) Harsten's theory was

____

(iii) Lynn and Joe Rapp, in their book, "Indoor Plants"say

____

(iv) Harsten said

____

(v) Harsten's experiment also proved

____

- that plants like classical music.
- that continuous sound of music does not allow breathing cells to be closed.
- that music soothes the plants.
- that the plants definitely react to music.
- that the humming sounds activate the breathing cells and make them stay open.

Remember:

The verbs used in the second parts of the sentences are in present simple/1st form because they talk about general/scientific facts.

Session - 7

7 Writing :

a. Provided below are some facts. Some of them are about Harsten and others, about Lynn and Rapp. But these are mixed. Put them neatly under two heads.

- Music helps plant grow.
- Wrote a book, 'Indoor Plants'.
- Plants respond to all sounds.
- Plants respond to only music.
- An engineer by profession
- Plants have definite likes and dislikes in music.
- Experiments with banana plants
- When listened to music plants grew faster.
- Plants like classical music.
- Plants dislike rock music.

Harsten

1. ______________________________
2. ______________________________
3. ______________________________
4. ______________________________
5. ______________________________

Lynn and Rapp

1. ______________________________
2. ______________________________
3. ______________________________
4. ______________________________
5. ______________________________

b. Using the hints write instructions for doing Harsten's experiment.

For Example: Take two banana plants.

- Give both plants the same ____________________ .
- Provide music to ____________________ .
- For about an ____________________ .

c. The following statements are not in order. See para- 4 and rearrange the statements in order as they come one after another in Harsten's experiment.

1. The plant grew faster.
2. Cells stay opened for longer periods.
3. Music stimulated plant cells.
4. The plant took more nutrient from air.

d. Based on the statements under two heads: under-1 about Harsten and under-2 Lynn and Rapp. Write two paras, one on Harsten and the other on Lynn and Rapp. Take the help of the statements under 7. a.

1. Harsten

____

____

____

____

____

2. **Lynn and Rapp**

____

____

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____

____

Session - 8

e. **Write answers to the following questions.**

(i) What is Harsten's theory ?

____

(ii) What did he use for his experiment ?

____

(iii) What did he give to both the banana plants ?

____

(iv) What was his finding ?

____

(v) Who are the authors of the book- "Indoor Plants" ?

____

(vi) What happened when the breathing cells of the plant were stimulated ?

____

(vii) What happened when the cells stayed open for longer periods ?

____

(viii) How is the finding of Lynn and Rapp different from that of Harsten ?

____

8 **Mental Talk :**

- Plants have definite likes and dislikes.
- Plants like classical music more.

9 **Let's Think :**

- Think of some new ways to know more about plants.
- Take up a project on Dr. Pranakrushna Parija, the great botanist of Odisha. Collect information about them from different sources.

FOLLOW- UP LESSON:

JAGDISH CHANDRA BOSE

Session - 1

I **PRE-READING**

- In this lesson you read about three scientists and their researches on plants.Read the following passage to know about Jagdish Chandra Bose, another scientist of India and his experiments.

II **WHILE-READING**

TEXT

- Jagdish Chandra Bose is one of the most prominent scientists of the world, who proved by experimentation that both animals and plants share much in common. He demonstrated that plants are also sensitive to heat, cold, light, noise and various other external stimuli. He made an instrument called Crescograph for the purpose. He wrote two very famous books – 'Response in the Living and Non-living' (1902) and "the Nervous Mechanism of Plants" (1926).

He was born on 30 November, 1858 at Mymensingh, now in Bangladesh. He got his elementary education from a vernacular school, because his father thought that Bose should learn his own mother tongue, Bengali, before studying a foreign language like English. Bose attended Cambridge after studying Physics at Calcutta University. He returned to India in 1884 after completing a B.Sc. degree from Cambridge University. He died in the year 1937.

- **Comprehension Question:**

1. Who is Jagdish Chandra Bose ?
2. Who shares much in common ?
3. How are plants similar to animals ?
4. What is a crescograph used for ? (sentence 2 of paragraph 1)
5. Name the books Jagdish Chandra Bose wrote ?
6. What are these books about ?
7. Where was he born ? When ?
8. Where did he have his primary education ? Why ?
9. What did he study at Calcutta University ?
10. Where did he get his B.Sc degree from ?
11. When did he die ?

Session - 2

III POST-READING

1 Vocabulary :

Match the words in A with their meanings in B

A	B
demonstrate	a tool used for doing a particular job
elementary	famous
experiment	several, different
instrument	Primary
prominent	the language spoken in a particular area, regional language
sensitive	scientific test or trial
stimulus (singular)	something that causes reaction
various	of outside
vernacular	to prove, to show clearly that something is true
external	showing reaction or response to show

2 Usage :

Read the following sentence.

He wrote two very famous books : "Response in the Living and Non-Living" and "the Nervous Mechanism of Plants".

The verb 'wrote' is used in its second form or past form- 'wrote'to state that the writer of the book is not alive .

Let's look at the following sentence:

Prativa Ray has written 'Silapadma'

In these sentences, we find the present perfect form of the verb 'write' has +written) shows that the author/writer is alive.

Rewrite the following sentences using the verb 'write' in its right form – past simple (wrote) or present perfect (has written).

(i) Jawaharlal Nehru (write) 'The Discovery of India'. (not alive)

Ans.____

(ii) Anita Desai (write) 'Cry the Peacock'. (alive)

____

(iii) Gopinath Mohanty (write) 'Paraja'. (not alive)

____

(iv) Sarala Das (write) 'The Odia Mahabharat'. (not alive)

____

(v) Chetan Bhagat (write) 'What Young India Wants'. (alive)

____

Session - 3

3 Writing :

a. Write answers to the following questions:

(i) Who was Jagadish Chandra Bose?

Ans. ____

____

(ii) What is the finding of his experiment?

Ans. ____

____

(iii) What is a Crescograph used for?

Ans. ____

____

(iv) Name the books he wrote

Ans. ____

____

b. Fill in this BIO-DATA of Jagadish Chandra Bose given below.

1. Name :
2. Profession :
3. Birth Place :
4. Date of Birth :
5. Primary Education :
6. Educational Institutions Attended
   a. Name :
      Subject :
   b. Name :
      Subject :
7. Language Known :
8. Experimented on:
9. Books he wrote :
10. The year of his passing away :

c. Write a paragraph on Sir Jagdish Chandra Bose using the information from the Bio-data you have already filled in.

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WORD NOTE

(The words / phrases have been defined mostly on contextual meanings.)

breathing cell (n)	-	cells that help breathing, ଶ୍ୱାସଗ୍ରନ୍ଥି
carried out (carry out)	-	did an experiment, (କାର୍ଯ୍ୟ ଟି) ସମ୍ପାଦିତ କଲେ
elementary (adj)	-	primary, ପ୍ରାଥମିକ ଶିକ୍ଷା
experiment (n)	-	scientific test, ବୈଜ୍ଞାନିକ ପରୀକ୍ଷା
high pitched (adj)	-	high volume (sound), ଅଧିକ ସାଉଣ୍ଡ, ଖୁବ୍ ଜୋର୍‌ରେ
nutrients (n)	-	rich elements of food that keep plants healthy, ପୁଷ୍ଟିକର
prominent (adj)	-	famous, ପ୍ରସିଦ୍ଧ, ବିଶିଷ୍ଟ, ବିଖ୍ୟାତ
scientific facts (n)	-	facts based on scientific researches, ବିଜ୍ଞାନ ସମ୍ମତ ତଥ୍ୟ
scientific way (n)	-	following scientific procedures, ବୈଜ୍ଞାନିକ ପଦ୍ଧତି / ପ୍ରଣାଳୀ
vibration, (n)	-	shaking, କମ୍ପନ
vernacular school	-	mother tongue medium school, ମାତୃଭାଷା ମାଧ୍ୟମରେ ଶିକ୍ଷାଦାନ ଦିଆଯାଉଥିବା ବିଦ୍ୟାଳୟ